

UNIVERSITAT AUTÒNOMA DE BARCELONA
UNIVERSITAT DE LES ILLES BALEARS
UNIVERSIDAD PÚBLICA DE NAVARRA

Master in Management, Organization and Business Economics
(MMOBE)

Common Ownership, Entry, and Innovation in a Cournot Duopoly

A thesis in partial fulfillment of the requirements
for the Master's degree in the MMOBE program,
co-organized by the Universitat Autònoma de Barcelona (UAB),
the Universitat de les Illes Balears (UIB), and the Universidad Pública de Navarra (UPNA),

Author:

Chunyan Wang

Advisors:

Eduard Alonso Paulí and Lluís Bru

Approval: ALONSO
PAULI
EDUARD -
DNI
46702528Q

Firmado
digitalmente por
ALONSO PAULI
EDUARD - DNI
46702528Q
Fecha: 2026.08.22
11:08:07 +02'00'

BRU
MARTINEZ
LLUIS - DNI
35074605L

Firmado digitalmente da BRU
MARTINEZ LLUIS - DNI 35074605L
ND: c=ES, o=UNIVERSITAT DE LES
ILLES BALEARS, ou=CERTIFICADO
ELECTRONICO DE EMPLEADO
PUBLICO,
serialNumber=IDCES-35074605L,
sn=BRU MARTINEZ,
givenName=LLUIS, cn=BRU
MARTINEZ LLUIS - DNI 35074605L
Data: 2026.08.24 09:40:14 +02'00'

September 2026

Abstract

This thesis studies how common ownership among competing firms affects entry and innovation in a Cournot duopoly. We first show that common ownership confined to incumbents attracts entry by an independent outsider, the opposite of models where the entrant is a member in the ownership network. We then consider a cost-reducing innovation with technological spillovers. When spillovers are strong, common ownership encourages investment by internalizing the rival's cost saving; when spillovers are weak and common ownership is high, the investment drives the rival out, and the efficiency gain leads to market concentration. These results show that the competitive consequences of common ownership, and therefore its effects on consumer surplus and global welfare, depend on the ownership structure and the technological environment.

Keywords: common ownership, entry, innovation, R&D spillovers, Cournot duopoly

1. Introduction and Brief Literature Review

1.1. What Is Common Ownership?

Common ownership refers to a situation in which the same investor simultaneously holds shares in multiple competing firms within the same industry (Ederer and Tecu 2025). Rather than assuming that a firm maximizes only its own profit, the common ownership hypothesis holds that a firm's manager maximizes a weighted sum of its shareholders' profits, including the interests those shareholders have in other firms (Ederer and Tecu 2025). This idea dates back to at least Rotemberg (1984), but it has only become an antitrust concern in recent decades as common ownership among competing firms has risen sharply: the average pairwise "profit weight" that S&P 500 firms place on each other's profits, as implied by the common ownership hypothesis, more than tripled between 1980 and 2017 (Ederer and Tecu 2025). The so-called "Big Three" asset managers (BlackRock, Vanguard, and State Street) hold large, overlapping equity stakes in publicly listed companies across many industries in the United States and globally, and common ownership continues to rise around the world (Antón et al. 2025b). This ownership structure has attracted considerable attention from antitrust enforcement agencies and academics alike: when the same investor owns stakes in rival firms, do those firms tend to compete less aggressively? The U.S. Federal Trade Commission (FTC) and Department of Justice (DOJ) answered this question affirmatively enough to explicitly identify common ownership as a potential source of competitive harm in the 2023 Merger Guidelines (U.S. Department of Justice and Federal Trade Commission 2023), and in 2024 several U.S. states sued major institutional investors, alleging that their common ownership of coal producers had anticompetitive effects (Ederer and Tecu 2025).

1.2. The Possible Effects of Common Ownership

A growing empirical literature asks whether common ownership measurably softens competition, and the evidence generally points toward higher prices in the industries studied by scholars. Azar, Schmalz, and Tecu (2018) provide the first such evidence for the U.S. airline industry: airline fares

are higher on routes and in quarters where common ownership among competing airlines, measured using a modified Herfindahl-Hirschman Index (MHHI) that incorporates ownership and control shares, is more pronounced, and accounting for common ownership raises measured market concentration by roughly 2,000 HHI points on average. The Herfindahl-Hirschman Index (HHI) is the standard measure of market concentration, computed as the sum of squared market shares per firm; the modified HHI (MHHI) used here augments it by also weighting rivals' market shares by the common ownership "profit weights" between firms, so a rise in MHHI with respect to HHI (holding market shares fixed) captures concentration attributable specifically to common ownership.¹ A 2,000-point rise is large by antitrust standards: the U.S. Merger Guidelines treat a market with an HHI above 2,500 as "highly concentrated," and consider a merger that raises HHI by more than 100–200 points in such a market to be presumptively anticompetitive. In other words, the concentration increase attributable to common ownership in the airline industry is roughly 10 to 20 times larger than the threshold that would trigger antitrust scrutiny for a merger. Applying a similar approach to banking, Azar, Raina, and Schmalz (2022) find that increased common ownership among banks is associated with lower deposit rates, consistent with softer competition for depositors.

These industry-specific findings raise the question of how widespread such effects are across the economy. Boller and Scott Morton (2020) address this question by using abnormal stock returns as a proxy for expected future profits: when a new entrant to the S&P 500 raises common ownership among competing firms, abnormal returns rise as well, and by more the larger the resulting increase in common ownership, evidence that the competitive effects of common ownership are not confined to a handful of well-studied industries. At the same time, these price-based findings remain contested: critics (Dennis, Gerardi, and Schenone 2022; Ederer and Tecu 2025) argue that the MHHI is mechanically related to market shares. That is, a firm that lowers its price gains market share, which mechanically raises its measured weight in the MHHI even if the underlying degree of common ownership has not changed; conversely, if common ownership

¹ Specifically, if the Herfindahl index measure of concentration is defined as $HHI = \sum_i s_i^2$, where s_i are market shares, the modified Herfindahl index is defined as $MHHI = HHI + \sum_i \sum_{j \neq i} \kappa_{ij} s_i s_j$, where s_j are market shares of firms other than firm i and κ_{ij} is the profit weight firm i places on firm j 's profits.

softens competition and raises prices, that price change itself shifts market shares and so mechanically moves the MHHI, apart from any change in ownership structure. Because market shares respond to prices in both directions, a regression of prices on the MHHI cannot cleanly separate the causal effect of common ownership, creating a feedback loop that Azar et al. (2018) only partially address, according to these critiques.

Beyond prices in a given market structure, common ownership can also affect the set of firms that compete with one another, and the intensity with which firms invest in cost-reducing and product-improving innovation. These two margins, entry and innovation, are the central focus of this thesis, and we discuss the relevant literature on each of them in turn, together with the positioning of this thesis in this literature.

1.3. The Effects of Common Ownership on Entry

Most theoretical models of common ownership, including the price-effect studies discussed above, take the number of active firms as fixed and do not consider how common ownership affects entry decisions. This omission matters because, if common ownership induces incumbents to raise prices and reduce output, the profit expectations of prospective entrants increase, and their entry decisions may change; entry then feeds back into market prices and consumer welfare. Sato and Matsumura (2020) provide the first theoretical treatment of this question: in a circular-market model, they show that common ownership reduces equilibrium entry because potential entrants themselves internalize the business-stealing losses they impose on co-owned incumbents. Newham, Seldeslachts, and Banal-Estañol (2025) document a closely related pattern empirically: in the pharmaceutical industry, higher common ownership between generic and branded drug manufacturers is associated with a lower likelihood of generic entry. Importantly, both studies share a common feature: the potential entrant is itself part of the common ownership network, so its own entry decision internalizes some of the harm it would inflict on incumbents.

A key theoretical question, therefore, first raised by Sato and Matsumura (2020), is whether this deterrence result is robust to alternative ownership configurations, in particular, to the case in which the potential entrant is not part of the common ownership network. Understanding the full

competitive effects of common ownership therefore requires a framework that explicitly incorporates endogenous entry and distinguishes between these different ownership structures. This is the gap that the entry analysis of this thesis addresses.

1.4. The Effects of Common Ownership on Innovation

Theoretically, López and Vives (2019) study cost-reducing R&D investment with technological spillovers in a Cournot oligopoly with overlapping ownership. They show that common ownership causes firms to internalize part of their rivals' profits and that, when spillovers are sufficiently strong, higher common ownership raises both innovation and output. However, how this result impacts on welfare depends critically on the welfare standard used. Under a total-surplus standard, common ownership may be welfare-enhancing because the increase in firm profits from softer competition can outweigh the loss in consumer surplus. Under a consumer-surplus welfare standard, meanwhile, common ownership is rarely desirable: even in relatively unconcentrated markets, where competition among firms is otherwise strong, the innovation gains from internalizing spillovers are not large enough to offset the price increase that common ownership causes, so a regulator maximizing consumer surplus would still prefer lower common ownership even when the underlying product market is not very concentrated. Extending this framework to allow for common ownership across the entire economy, product differentiation, and firm-varying spillovers, Antón et al. (2025a) provide empirical evidence for two offsetting channels: common ownership between closer product-market competitors reduces innovation, whereas common ownership between firms that are closer in technology space increases it, so that the net effect of common ownership on innovation depends on how the similarity of firms in their products relative to the technology they use.

The broader empirical literature reinforces this picture of genuine ambiguity. Using data on common venture-capital ownership of pharmaceutical startups, Li, Liu, and Taylor (2023) find that common ownership improves overall innovation efficiency, even though common venture-capital owners also actively hold back weaker projects and redirect funding away from lagging startups; that is, the aggregate efficiency gain partly reflects a reallocation of resources rather than a uniform increase in innovative activity across all portfolio firms. Studying institutional blockholders more

broadly, He and Huang (2017) similarly find that common ownership is, on average, associated with higher corporate innovation productivity. Kostovetsky and Manconi (2020) exploit the exogenous increase in shared institutional ownership generated by index additions and show that a newly added firm's patents subsequently receive more citations, consistent with common ownership facilitating the diffusion of innovation across firms. In contrast, Chiao, Qiu, and Wang (2021) find that common ownership is, on average, negatively related to patent grants, patent citations, and R&D expenditures, although it also reduces the likelihood that commonly owned firms become involved in patent litigation and speeds up settlement when disputes do arise, suggesting that common ownership can still mitigate hold-up problems between firms with complementary patent portfolios even as it dampens measured innovation output. Reconciling these divergent findings, Borochin, Yang, and Zhang (2020) show that whether common ownership increases or decreases patent output and citations depends on the type of institutional owner generating the common ownership link, implying that the innovation consequences of common ownership are heterogeneous across the investor base rather than uniform.

As we said above, taken together this literature establishes that the innovation effects of common ownership are genuinely ambiguous, depending on the strength of technological spillovers, the type of common owner, and whether firms are product-market or technology-space rivals. Like the entry literature discussed before, however, existing theoretical treatments of common ownership and innovation, including López and Vives (2019) and Antón et al. (2025a), take the number of active firms as fixed: they study how common ownership shapes R&D incentives given that both firms remain in the market, but not whether the cost asymmetry created by innovation itself can push a commonly owned rival out of the market altogether. The innovation extension of this thesis addresses this second gap. Abstracting from entry, as in the existing literature, it asks how common ownership shapes one incumbent's incentive to undertake cost-reducing R&D with technological spillovers, and it shows that, when spillovers are weak and common ownership is high, the investment can drive the less efficient rival out of the market entirely, turning a cost-reducing innovation into a device for market concentration.

1.5. This Thesis

This paper first develops a Cournot duopoly model with common ownership (Section 2) and then extends it in two directions:

- First, it introduces endogenous entry to study the effects of common ownership on market competition and welfare. The model builds on the framework of López and Vives (2019): market demand is linear, given by $P = 1 - Q$, and all firms have the same constant marginal cost $c \in (0, 1)$. Common ownership between the two incumbent firms is captured by the parameter $\alpha \in [0, 1]$: rather than maximizing its own profit π_i alone, the manager of firm i maximizes $\pi_i + \alpha\pi_j$ (where $j \neq i$), that is, the manager of firm i partially internalizes the rival's profit. Crucially, the potential entrant does not have common ownership with either incumbent: it maximizes only its own net profit and pays a sunk entry cost F_E to enter. The game proceeds in two stages: in the first stage, the potential entrant decides whether to enter; in the second stage, all active firms simultaneously choose quantities. This configuration is the complement of the setting studied by Newham et al. (2025) and Sato and Matsumura (2020), where the entrant is itself part of the common ownership network.
- The thesis also extends the baseline framework in a second model that studies innovation rather than entry. Holding the market structure fixed at the two commonly owned incumbents and abstracting from entry, firm 1 can undertake a cost-reducing investment that lowers its own marginal cost from c to 0 at a sunk cost F_I . Because of technological spillovers, the same investment also lowers the rival's marginal cost from c to $(1 - m)c$, where $m \in [0, 1]$ indexes the strength of the spillover. The game again has two stages: firm 1 decides whether to invest, and then both active firms compete in quantities given their realized costs. This second model isolates the innovation margin of common ownership, asking how the interaction between the spillover parameter m and the ownership parameter α shapes both the incentive to invest and, crucially, whether the investment leaves the rival active or drives it out of the market.

Solving both extensions of the basic model yields the following main results.

First, common ownership raises the entrant's expected profit and thereby encourages entry. When entry occurs, the entrant's equilibrium output is $q^E(\alpha) = (1 + \alpha)(1 - c) / [2(2 + \alpha)]$, and its gross profit is $[q^E(\alpha)]^2$. This profit is monotonically increasing in α : because common ownership leads

incumbents to reduce their output and raise the market price, the entrant faces a larger residual demand. Given entry cost F_E , there exists a unique threshold $\bar{\alpha}(F_E) = [4\sqrt{F_E} - (1 - c)] / [(1 - c) - 2\sqrt{F_E}]$ such that entry occurs when $\alpha \geq \bar{\alpha}$, and entry is blocked when $\alpha < \bar{\alpha}$. In other words, sufficiently high common ownership attracts entry, while low levels of common ownership may deter it. The sign reversal relative to Sato and Matsumura (2020) arises because, in our model, the entrant's objective function does not include the incumbents' profits: the business-stealing internalization channel that deters entry in their framework is absent here.

Second, regardless of whether entry occurs, common ownership reduces total output and consumer surplus. In the no-entry case, total output is $Q_0(\alpha) = 2(1 - c) / (3 + \alpha)$, which decreases as α rises (falling from the standard Cournot level of $2(1 - c)/3$ when $\alpha = 0$ to the monopoly output of $(1 - c)/2$ when $\alpha = 1$). In the entry case, total output is $Q_1(\alpha) = (3 + \alpha)(1 - c) / [2(2 + \alpha)]$, which also decreases in α but is always greater than $Q_0(\alpha)$. At the threshold $\bar{\alpha}$, total output and consumer surplus exhibit a downward jump if α is reduced: once common ownership falls below the level needed to attract entry, the market shifts from three firms to two, causing output to drop sharply and welfare to deteriorate. The occurrence of entry can offset the anticompetitive effects of common ownership. From the perspective of consumer welfare, there is therefore an “optimal” range of α : a level of common ownership that is just high enough to induce entry but not so high as to drive output back down.

Third, in the innovation extension, the effect of common ownership on firm 1's incentive to invest in cost-reducing R&D is non-monotone in α , turning from increasing to decreasing exactly when the investment drives the less efficient rival out of the market. Two opposing channels are at work: a dampened business-stealing channel, through which common ownership reduces the private gain from stealing market share from the rival and so weakens the incentive to invest, and an internalized spillover channel, through which common ownership makes firm 1's manager partly value the cost saving that the investment confers on the rival, strengthening the incentive to invest, refining the logic of López and Vives (2019). Crucially, when the spillover is weak and common ownership is high enough, the investment lowers firm 1's cost by more than it lowers the cost of the rival, and the resulting cost asymmetry drives the less efficient rival out of the market entirely: common ownership then turns a cost-reducing investment into a device for concentrating the market in the hands of the efficient incumbent. Consumers always weakly benefit from the

investment relative to the same level of common ownership without investment, but for $c < 1/4$ they are worse off than under ordinary Cournot competition with no common ownership even if it results in no innovation.

The remainder of this paper is structured as follows. Section 2 presents the baseline Cournot duopoly model with common ownership and derives the equilibrium that serves as the benchmark for both extensions. Section 3 establishes the entry model (without R&D), solves the equilibrium in both the no-entry and entry cases, derives the entry threshold, and conducts welfare comparisons. Section 4 introduces R&D spillovers, analyzing how common ownership affects the incentive to invest and how it interacts with the market-structure effect of driving the less efficient rival out of the market. Section 5 concludes the paper: it draws out the policy implications of the model, discusses its limitations, and outlines possible directions for future research.

2. The Baseline Model

Consider a market served by two incumbent firms, indexed $i = 1, 2$, and Market demand is characterised by the linear inverse demand function

$$P = 1 - Q, \quad (1)$$

where Q denotes total industry output. All firms have constant marginal cost normalised to c , with $0 < c < 1$. Firms compete in quantities under the Cournot (1838) model, in which each firm simultaneously chooses output taking rivals' quantities as given. The linear inverse demand in equation (1) is a particular case, when products are not differentiated, of Singh and Vives (1984), whose duopoly framework provides the microeconomic foundation for this specification.

The two incumbents are linked by a common ownership parameter $\alpha \in [0, 1]$, capturing the degree to which each incumbent's manager internalizes the profit of the rival incumbent. Note that while the baseline model uses $\alpha \in [0, 1]$, the innovation model in Section 4 restricts to $\alpha \in [0, 1)$, because the equilibrium quantities derived in this section contain a denominator $(1-\alpha)$, excluding $\alpha = 1$ keeps the duopoly equilibrium well-defined. Specifically, the manager of incumbent i maximizes

$$\Phi_i = \pi_i + \alpha \pi_j, j \neq i, \quad (2)$$

where $\pi_i(q_i, q_j) = (P(q_i + q_j) - c)q_i$ is the individual profit of firm i . At $\alpha = 0$ each firm

maximizes its own profit (standard Cournot); at $\alpha = 1$ each manager fully internalizes the rival's profit, which is equivalent to joint-profit (cartel) maximization. Following López and Vives (2019), α summarizes the degree of profit internalization arising from overlapping ownership structures such as common institutional ownership or cross-shareholding.

To simplify, along the Master Thesis we assume that α is the same for both incumbent firms. Moreover, we abstract from agency conflicts between managers and shareholders: managers are assumed to maximize the objective in equation (2) directly, and α is treated as an exogenous parameter determined by the ownership structure. The parameter α therefore summarizes the competitive consequences of the ownership structure without requiring a full specification of the underlying corporate governance mechanism.

It is important to understand what drives the magnitude of α in practice. A low value of α can arise from two distinct situations: (i) the level of common ownership itself is low, or (ii) the level of common ownership is high, but shareholders have limited influence over managerial decisions. Conversely, a high value of α requires both a substantial ownership stake and the capacity to influence managerial decisions. Crucially, a “substantial” stake does not necessarily mean a very high percentage of shares: in firms with a large free float, even a relatively modest stake can have significant influence over management, and thus generate a high α .

We extend the baseline model to incorporate a firm E that eventually may enter the market, to study the entry issue on common ownership in Section 3, and to incorporate an incumbent firm that can undertake a cost-reducing innovation in Section 4.

3. Common ownership and entry

3.1. Setup

The strategic interaction is modelled as a two-stage game. In Stage 1, firm E decides whether to enter. If the entrant chooses to enter, it earns market profit $\pi_E(q_1, q_2, q_E) = (P(q_1 + q_2 + q_E) - c)q_E$, defined analogously to incumbent profits, but also bears a sunk entry cost $F_E > 0$. Its net profit is therefore $\Pi^E = \pi_E - F_E$, and entry is profitable if and only if $\Pi^E \geq 0$.

In Stage 2, all active firms simultaneously choose output quantities. The properties of the Cournot model used here, the existence and uniqueness of a symmetric interior equilibrium, strategic substitutability of quantities, and the comparative statics of concentration are discussed in detail in Tirole (1988, chaps. 5–6) and Vives (1999). The appropriate equilibrium concept for a finite, sequential game of this kind is the Subgame-Perfect Nash Equilibrium (SPNE), which requires that strategies form a Nash Equilibrium (NE) in every subgame. We solve the model by backward induction: we first derive the Stage 2 Nash Equilibrium in quantities for each possible market structure (two incumbents only, or two incumbents plus the entrant), then use the resulting equilibrium profits to determine the Stage 1 entry decision.

- **Stage 1 (entry decision):** Firm E enters if and only if $\Pi^E(\alpha, F_E) \geq 0$, where Π^E denotes the net profit from entry.
- **Stage 2 (quantity competition):** All active firms simultaneously choose quantities. Each incumbent i maximizes Φ_i ; the entrant (if present) maximizes π_E .

This sequential structure is a standard and natural representation of market interactions in industrial organization (Tirole 1988; Vives 1999). Entry decisions involve irreversible commitments, such as building production capacity, establishing supplier relationships, and obtaining regulatory approvals that must be made before firms interact in the product market. Quantity competition therefore takes place after entry is settled, making the two-stage game a fair description of how firms interact in markets with potential entry.

3.2. No-Entry Equilibrium

We first characterize the Stage 2 equilibrium when firm E remains outside the market. With total output $Q = q_i + q_j$ and profit $\pi_i = (1 - q_i - q_j - c)q_i$, the objective function of incumbent i is

$$\Phi_i = (1 - q_i - q_j - c)q_i + \alpha(1 - q_i - q_j - c)q_j \quad (3)$$

The first-order condition (FOC) with respect to q_i is

$$\partial \Phi_i / \partial q_i = (1 - c - 2q_i - q_j) - \alpha q_j = 0. \quad (4)$$

Re-arranging yields the best-response function of incumbent i :

$$q_i = BR_i(q_j) = \frac{1-c}{2} - \frac{1+\alpha}{2} q_j. \quad (5)$$

Note that quantities are strategic substitutes in this model, which means that an increase in one firm's output leads rivals to reduce theirs, it's a property formally established by Bulow, Geanakoplos, and Klemperer (1985). Compared with the standard Cournot best response ($\alpha = 0$), a positive α steepens the slope from $-1/2$ to $-(1 + \alpha)/2$. Internalizing part of the rival's profit induces each manager to contract output more aggressively in response to any expansion by the rival, effectively shifting the best-response function inward. Imposing symmetry at equilibrium ($q_0 \equiv q_1^* = q_2^*$) and solving:

$$q_0(\alpha) = \frac{1-c}{3+\alpha}, \quad Q_0(\alpha) = \frac{2(1-c)}{3+\alpha}. \quad (6)$$

Differentiating total output with respect to α :

$$\frac{dQ_0}{d\alpha} = -\frac{2(1-c)}{(3+\alpha)^2} < 0. \quad (7)$$

Output is strictly decreasing in α . At $\alpha = 0$ we recover the standard symmetric Cournot duopoly outcome, $Q_0(0) = 2(1 - c)/3$; at $\alpha = 1$ we obtain the monopoly output, $Q_0(1) = (1 - c)/2$. The remaining equilibrium values are

$$\text{Price: } P_0(\alpha) = 1 - Q_0(\alpha) = \frac{1+\alpha+2c}{3+\alpha}, \quad (8)$$

$$\text{Each incumbent's profits: } \pi_0(\alpha) = q_0(\alpha) \cdot (P_0(\alpha) - c) = \frac{(1+\alpha)(1-c)^2}{(3+\alpha)^2}, \quad (9)$$

$$\text{Consumer Surplus: } CS_0(\alpha) = \frac{Q_0(\alpha)^2}{2} = \frac{2(1-c)^2}{(3+\alpha)^2}, \quad (10)$$

$$\text{Total Welfare: } TW_0(\alpha) = CS_0(\alpha) + 2\pi_0(\alpha) = \frac{2(\alpha+2)(1-c)^2}{(3+\alpha)^2}. \quad (11)$$

These four measures respond to α in different directions. The equilibrium price $P_0(\alpha)$ is strictly increasing in α , since $dP_0/d\alpha = 2(1 - c)/(3 + \alpha)^2 > 0$. Incumbent profit $\pi_0(\alpha)$ is non-decreasing in α : $d\pi_0/d\alpha = (1 - \alpha)(1 - c)^2/(3 + \alpha)^3 \geq 0$ for all $\alpha \in [0, 1]$, with equality only at $\alpha = 1$. By contrast, consumer surplus $CS_0(\alpha)$ ($dCS_0/d\alpha = -4(1 - c)^2/(3 + \alpha)^3 < 0$) and total welfare $TW_0(\alpha)$ ($dTW_0/d\alpha = -2(1 + \alpha)(1 - c)^2/(3 + \alpha)^3 < 0$) are both strictly decreasing in α . Higher common ownership therefore produces a shift of rents from consumers to producers: output falls, the price rises, incumbents capture higher profits, but the deadweight loss from the output contraction grows with

α , causing overall welfare to decline. As α rises, each incumbent internalizes a greater share of its rival's profit and therefore has an incentive to restrain its own output, since doing so raises the market price and benefits the rival. This mutual output restraint drives the equilibrium price upward, from the competitive Cournot level at $\alpha = 0$ toward the joint-profit-maximizing (monopoly) level at $\alpha = 1$.

The mechanism through which common ownership translates into higher prices and lower output has been empirically documented across several industries. Antón, Ederer, Giné, and Schmalz (2023) show that the channel operates via managerial incentives: common ownership makes performance-sensitive pay contracts less optimal for shareholders who also hold stakes in rivals, because aggressive cost-reduction and output expansion by one firm reduces the profits of other portfolio firms. As a result, managers in industries with higher common ownership face flatter incentive schemes, exert less effort, and their firms produce less and charge higher prices. Ederer and Tecu (2025) further document that these anticompetitive effects of common ownership are present across a wide range of industries and market settings, making it a first-order concern for antitrust enforcement.

3.3. Equilibrium with Entry

Suppose now that firm E has entered, so three firms are active. Total output is $Q = q_i + q_j + q_E$. The incumbent managers maximize

$$\Phi_i = (1 - c - q_i - q_j - q_E)q_i + \alpha(1 - c - q_i - q_j - q_E)q_j \quad (12)$$

while the entrant maximizes $\Phi_E = (1 - c - q_i - q_j - q_E)q_E$. Crucially, α does not appear in the entrant's objective: the entrant does not internalize the incumbents' profits. The three FOCs are

$$(i): 1 - c - 2q_i - (1 + \alpha)q_j - q_E = 0, \quad (13)$$

$$(j): 1 - c - (1 + \alpha)q_i - 2q_j - q_E = 0, \quad (14)$$

$$(E): 1 - c - q_i - q_j - 2q_E = 0. \quad (15)$$

Imposing symmetry between the two incumbents at equilibrium ($q_i^* = q_j^* \equiv q_I^*$) and solving the resulting two-equation system:

$$q_1^I(\alpha) = \frac{1-c}{2(2+\alpha)}, \quad (16)$$

$$q_1^E(\alpha) = \frac{(1+\alpha)(1-c)}{2(2+\alpha)}, \quad (17)$$

$$Q_1(\alpha) = \frac{(3+\alpha)(1-c)}{2(2+\alpha)}. \quad (18)$$

Several notable results follow. First, total output is again strictly decreasing in α :

$$\frac{dQ_1}{d\alpha} = -\frac{1(1-c)}{2(2+\alpha)^2} < 0. \quad (19)$$

At $\alpha = 0$ we obtain the standard three-firm Cournot outcome, $Q_I(0) = 3(1-c)/4$; at $\alpha = 1$, $Q_I(1) = 2(1-c)/3$. Notably, $Q_I(1) = Q_0(0) = 2(1-c)/3$, a benchmark identity used in the welfare comparison of Section 3.5.

Second, the entrant's equilibrium quantity is strictly increasing in α :

$$\frac{dq_1^E}{d\alpha} = \frac{1(1-c)}{2(2+\alpha)^2} > 0. \quad (20)$$

As α rises, the incumbents collectively reduce their aggregate output $2q_1^I(\alpha) = (1-c)/(2+\alpha)$, leaving larger residual demand for the entrant. Higher common ownership among incumbents thus indirectly benefits the independent entrant by softening competitive pressure.

Third, the price-minus-cost margin equals the entrant's own equilibrium quantity:

$$P_1(\alpha) - c = 1 - Q_1(\alpha) - c = \frac{(1+\alpha)(1-c)}{2(2+\alpha)} = q_1^E(\alpha). \quad (21)$$

This identity implies that the entrant's gross profit is simply the square of its equilibrium quantity:

$$\pi_1^E(\alpha) = q_1^E(\alpha) \cdot (P_1(\alpha) - c) = [q_1^E(\alpha)]^2 = \frac{(1+\alpha)^2(1-c)^2}{4(2+\alpha)^2}. \quad (22)$$

The gross profit $\pi_1^E(\alpha)$ is strictly increasing in α . The entrant's net profit is

$$\Pi^E(\alpha, F_E) = \frac{(1+\alpha)^2(1-c)^2}{4(2+\alpha)^2} - F_E. \quad (23)$$

The remaining equilibrium values with entry are

$$\pi_1^I(\alpha) = q_1(\alpha) \cdot (P_1(\alpha) - c) = \frac{(1+\alpha)(1-c)^2}{4(2+\alpha)^2}, \quad (24)$$

$$CS_1(\alpha) = \frac{Q_1(\alpha)^2}{2} = \frac{(3+\alpha)^2(1-c)^2}{8(2+\alpha)^2}, \quad (25)$$

$$TW_1(\alpha) = CS_1(\alpha) + 2\pi_1^I(\alpha) + \Pi^E(\alpha, F_E) = \frac{(3\alpha^2 + 14\alpha + 15)(1-c)^2}{8(2+\alpha)^2} - F_E. \quad (26)$$

3.4. The Entry Threshold

It is important to recall from Section 3.3 that the entrant's gross profit $\pi_1^E(\alpha)$ is strictly increasing in α (equation (22)). In Stage 1, firm E enters if and only if $\Pi^E(\alpha, F_E) \geq 0$. This entry decision framework, in which a potential entrant compares expected post-entry profits against a fixed sunk cost, follows the empirical entry literature of Bresnahan and Reiss (1991), who model entry as a threshold condition on market profitability. From equation (23), this is equivalent to

$$\frac{(1+\alpha)(1-c)}{2(2+\alpha)} \geq \sqrt{F_E}. \quad (27)$$

Solving for α yields the unique threshold

$$\bar{\alpha}(F_E) = \frac{4\sqrt{F_E} - (1-c)}{(1-c) - 2\sqrt{F_E}}. \quad (28)$$

The expression for $\bar{\alpha}(F_E)$ in equation (28) is well defined only when its denominator, $(1-c) - 2\sqrt{F_E}$, is strictly positive, i.e. when $F_E < [(1-c)/2]^2$. For $F_E \geq [(1-c)/2]^2$, the threshold either diverges or is negative, and entry is never profitable for any $\alpha \in [0, 1]$. In practice, since we restrict attention to $F_E \in [[(1-c)/4]^2, [(1-c)/3]^2]$, the denominator is strictly positive in this range. This threshold means that: if $\alpha \geq \bar{\alpha}(F_E)$, then the entrant's gross profit exceeds the sunk cost F_E and entry is profitable; if $\alpha < \bar{\alpha}(F_E)$, then the entrant's gross profit is smaller than F_E and the entrant stays out. If $F_E < [(1-c)/4]^2$, then $\bar{\alpha} < 0$ and entry always occurs; if $F_E > [(1-c)/3]^2$, then $\bar{\alpha} > 1$ and entry never occurs.

Proposition 1 summarizes the results of this analysis:

Proposition 1. For $F_E < [(1-c)/4]^2$ entry always occurs, for $F_E > [(1-c)/3]^2$ entry never occurs. For $F_E \in [[(1-c)/4]^2, [(1-c)/3]^2]$, there exists a unique threshold $\bar{\alpha}(F_E) \in [0, 1]$, defined by equation (28), such that: (a) entry occurs if and only if $\alpha \geq \bar{\alpha}(F_E)$; and (b) $\bar{\alpha}(F_E)$ is strictly increasing in F_E .

Proof. The gross profit $\pi_1^E(\alpha) = (1+\alpha)^2(1-c)^2/[4(2+\alpha)^2]$ is strictly increasing and continuous in α (equation (22)), so it crosses the level F_E exactly once for $F_E \in [\pi_1^E(0), \pi_1^E(1)] = [[(1-c)/4]^2,$

$[(1 - c)/3]^2$. That unique crossing defines $\bar{\alpha}(F_E)$. $\square$

The threshold $\bar{\alpha}(F_E)$ is strictly increasing in F_E . The key economic insight is that entry becomes more attractive as α rises, because the incumbents' mutual accommodation cedes residual demand to the entrant. This contrasts with the conventional intuition that common ownership deters entry. That intuition originates in settings where the same investor holds stakes in both the incumbents *and* the potential entrant, so that the entrant internalizes the profit losses its entry would impose on co-owned rivals, what Sato and Matsumura (2020) call the business-stealing internalization effect. In their circular-market model with free entry, this effect always dominates the pro-entry effect of higher markups, so common ownership unambiguously reduces entry.

Newham, Seldeslachts, and Banal-Estañol (2025) confirm this empirically in the pharmaceutical industry: when large institutional investors such as BlackRock and Vanguard hold stakes in both branded incumbents and generic potential entrants, a one-standard-deviation increase in common ownership reduces the probability of generic entry by 9 to 13 percent.

In our model, by contrast, α links only the two incumbents and the entrant have no common ownership with either, so α does not appear in the entrant's objective (equation (15)). The business-stealing channel is therefore absent, and a higher value of α instead raises the market price and expands residual demand, making entry more attractive. Our model and these two papers are thus complementary: the direction of common ownership's effect on entry depends critically on whose ownership is common. This distinction has direct antitrust implications, discussed below.

3.5. Welfare Comparison

We now combine the two regimes to characterize the subgame-perfect equilibrium of the full two-stage game and compare welfare outcomes.

Output, price and consumer surplus. For any given α , total output is strictly higher and the price strictly lower in the entry regime than in the no-entry regime:

$$Q_1(\alpha) - Q_0(\alpha) = \frac{(1+\alpha)^2(1-c)}{2(2+\alpha)(3+\alpha)} > 0, \quad (29)$$

with the same expression giving $P_0(\alpha) - P_I(\alpha) > 0$. Consequently $CS_I(\alpha) > CS_0(\alpha)$ for all $\alpha \in [0, 1]$: consumers always prefer the entry regime.

Equilibrium of $Q(\alpha)$ and $CS(\alpha)$. By backward induction, for $F_E \in [(1-c)/4]^2, [(1-c)/3]^2$:

- For $\alpha < \bar{\alpha}(F_E)$: no entry. $Q(\alpha) = Q_0(\alpha) = 2(1-c)/(3+\alpha)$, strictly decreasing in α .
- For $\alpha \geq \bar{\alpha}(F_E)$: entry occurs. $Q(\alpha) = Q_I(\alpha) = (3+\alpha)(1-c)/[2(2+\alpha)]$, strictly decreasing in α .

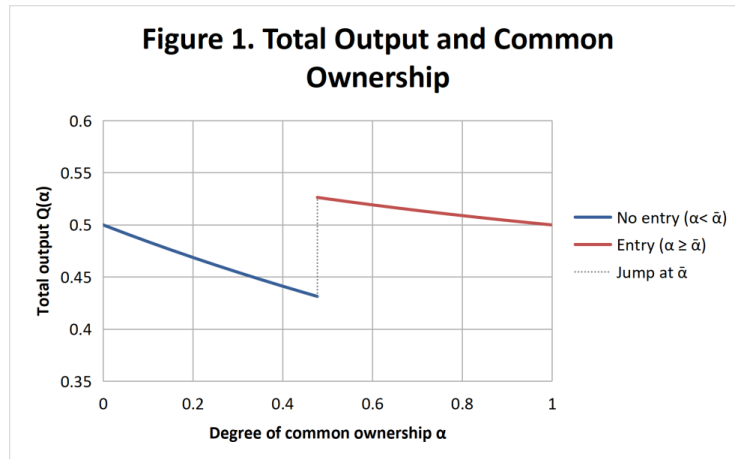
Since $Q_I(\alpha) > Q_0(\alpha)$ for all α , crossing the threshold $\bar{\alpha}$ from below triggers a discrete upward jump in total output. Within each regime output is continuously decreasing. The overall path $Q(\alpha)$ therefore takes the shape of two downward-sloping segments connected by an upward discontinuity at $\bar{\alpha}$. Consumer surplus $CS(\alpha) = Q(\alpha)^2/2$ inherits the same sawtooth shape.

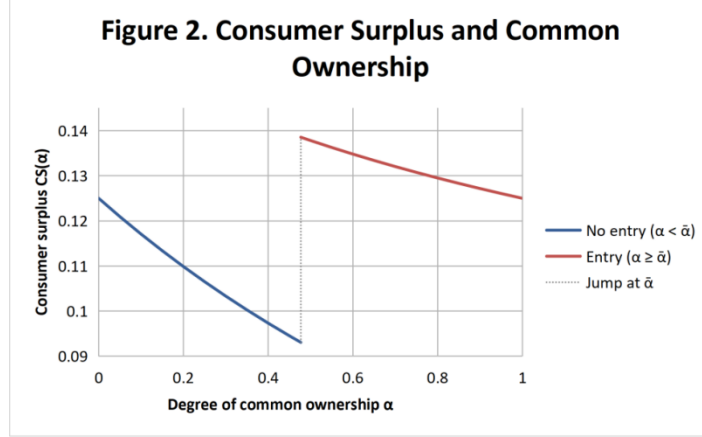
Benchmark. A useful cross-regime result is

$$Q_0(0) = Q_1(1) = \frac{2(1-c)}{3} \quad (30)$$

The no-entry output under pure Cournot competition ($\alpha = 0$) coincides with the entry-regime output under full profit internalization ($\alpha = 1$). The same holds for consumer surplus: $CS_0(0) = CS_I(1) = 2(1-c)^2/9$.

Figures 1 and 2 below illustrate this discussion.





Figures 1 and 2. Total output $Q(\alpha)$ and consumer surplus $CS(\alpha)$ as functions of the degree of common ownership α , for entry cost $F_E = 0.05$ and marginal cost $c = 1/4$ (entry threshold $\bar{\alpha} \approx 0.48$). Blue curves denote equilibrium values in the no-entry regime ($\alpha < \bar{\alpha}$); red curves denote equilibrium values in the entry regime ($\alpha \geq \bar{\alpha}$). Both $Q(\alpha)$ and $CS(\alpha)$ are strictly decreasing within each regime but jump upward at the threshold, reflecting the pro-competitive effect of entry.

Aggregate profits. Aggregate industry profit falls strictly upon entry:

$$\Pi_1(\alpha) = 2\pi_1^I(\alpha) + \pi_1^E(\alpha) < 2\pi_0(\alpha) = \Pi_0(\alpha) \quad \text{for all } \alpha \in [0, 1], \quad (31)$$

To see why, note that $\Pi_1(\alpha) - \Pi_0(\alpha) = (1 + \alpha)^2(1 - c)^2(\alpha^2 - 5)/[4(2 + \alpha)^2(3 + \alpha)^2]$. Since $\alpha \in [0, 1]$ the numerator factor $(\alpha^2 - 5)$ is strictly negative for all α in the relevant range, making the entire expression negative. The intuition is that the profit loss suffered by each incumbent, who now faces a third competitor and loses market share, more than offsets the gross profit earned by the entrant. Entry therefore represents a net loss for the industry, even though it benefits consumers through lower prices and higher output.

Finally, firm's profit satisfies $\pi_1^I(\alpha) < \pi_0(\alpha)$ for all α . The incumbents are therefore made strictly worse off by entry, even though it is their own mutual accommodation (induced by a higher α) that attracts the entrant in the first place.

Total welfare. Unlike the no-entry case, total welfare in the entry regime includes the sunk cost F_E as a social cost, since F_E represents real resources consumed by entry. From equations (11) and (26):

$$TW_1(\alpha) - TW_0(\alpha) = \frac{(1+\alpha)^3(3\alpha+7)(1-c)^2}{8(2+\alpha)^2(3+\alpha)^2} - F_E. \quad (32)$$

This difference is positive if and only if $F_E < (1+\alpha)^3(3\alpha+7)(1-c)^2/[8(2+\alpha)^2(3+\alpha)^2]$. Since the entry condition requires $F_E < \pi_I^E(\alpha) = (1+\alpha)^2(1-c)^2/[4(2+\alpha)^2]$, and since $\pi_I^E(\alpha) > (1+\alpha)^3(3\alpha+7)(1-c)^2/[8(2+\alpha)^2(3+\alpha)^2]$ for all $\alpha \in [0, 1]$, the entry condition alone does not guarantee $TW_I > TW_0$. Entry improves total welfare only when the sunk cost F_E is sufficiently small relative to the gross welfare gain. When F_E is large (but still below the entry threshold), entry occurs but dissipates enough surplus that total welfare may fall. This stands in contrast to consumer surplus, where entry always benefits consumers regardless of F_E .

We summarize the main distributional findings in the following proposition.

Proposition 2. For all $\alpha \in [0, 1]$: (a) $Q_I(\alpha) > Q_0(\alpha)$ and $P_I(\alpha) < P_0(\alpha)$: entry increases output and reduces price. (b) $CS_I(\alpha) > CS_0(\alpha)$: entry always improves consumer surplus. (c) $\pi_I^I(\alpha) < \pi_0(\alpha)$: entry reduces each incumbent's profit. (d) $TW_I(\alpha) > TW_0(\alpha)$ if and only if $F_E < (1+\alpha)^3(3\alpha+7)(1-c)^2/[8(2+\alpha)^2(3+\alpha)^2]$: entry improves total welfare only when the sunk cost is sufficiently small.

Taken together, Propositions 1 and 2 highlight a key asymmetry in the effects of common ownership: within any given market structure, higher α reduces output and consumer surplus; yet the same mechanism attracts entry, which always benefits consumers. Whether it also raises total welfare depends on how large the sunk entry cost F is relative to the gross surplus gain from a more competitive market structure.

4. Common Ownership and Innovation

4.1. Setup

Now we extend the baseline model to incorporate a cost-reducing innovation. The market structure remains as in Section 2: two incumbent firms indexed $i = 1, 2$, competing in quantities under the Cournot (1838) model with inverse demand $P = 1 - Q$, and linked by a common ownership parameter $\alpha \in [0, 1)$. Both firms initially have the same constant marginal cost $c \in (0, 1)$, and we set aside the entry decision studied in Section 3.

Firm 1 can undertake a cost-reducing investment that lowers its own marginal cost from c to 0, at a sunk cost $F_I > 0$. Because of technological spillovers, this investment also lowers the rival's marginal cost from c to $(1 - m)c$, where $m \in [0, 1]$ indexes the strength of the spillover: $m = 0$ means no spillover (firm 2 keeps cost c); $m = 1$ means full spillover (firm 2's cost also falls to 0). The rival's cost is therefore decreasing in m .

One modelling assumption underpin this setup: firms have asymmetric innovation capabilities: only firm 1 can undertake the cost-reducing investment, reflecting the realistic feature that firms differ in their technological capacity to innovate. Firm 2 cannot invest directly but can benefit indirectly through the spillover parameter m : a higher m means the technology diffuses more strongly to the rival, so even though firm 2 cannot innovate on its own, it may gain substantially from firm 1's investment when m is large.

The game has two stages:

- **Stage 1 (investment decision):** Firm 1 decides whether to invest, incurring sunk cost F_I if it does so.
- **Stage 2 (quantity competition):** Both active firms simultaneously choose quantities given their marginal costs.

We solve by backward induction, deriving the Stage 2 equilibrium for arbitrary marginal-cost pairs (c_1, c_2) first, then using those payoffs to evaluate the Stage 1 investment decision. Throughout, the manager of firm i maximizes

$$\Phi_i = \pi_i + \alpha \pi_j, j \neq i, \quad (33)$$

where $\pi_i = (P - c_i)q_i = (1 - c_i - q_1 - q_2)q_i$ is the individual profit of firm i .

4.2. Preliminary: The Quantity Game with Asymmetric Costs

We first solve Stage 2 in full generality for arbitrary marginal costs (c_1, c_2) . The objective function of firm i is

$$\Phi_i = (1 - c_i - q_1 - q_2)q_i + \alpha(1 - c_j - q_1 - q_2)q_j. \quad (34)$$

Differentiating with respect to q_i and setting equal to zero yields the first-order condition

$$1 - c_i - 2q_i - q_j - \alpha q_j = 0, \quad (35)$$

which gives the best-response function

$$q_i = \frac{1-c_i}{2} - \frac{1+\alpha}{2} q_j. \quad (36)$$

Solving the two first-order conditions simultaneously (the determinant of the system is $D = (1 - \alpha)(3 + \alpha) > 0$ for $\alpha \in [0, 1)$) yields the unique interior equilibrium:

$$q_1^* = \frac{2(1-c_1) - (1+\alpha)(1-c_2)}{(1-\alpha)(3+\alpha)}, q_2^* = \frac{2(1-c_2) - (1+\alpha)(1-c_1)}{(1-\alpha)(3+\alpha)}. \quad (37)$$

Both quantities are strictly positive if and only if each numerator in equation (37) is positive, i.e. $2(1 - c_i) > (1 + \alpha)(1 - c_j)$ for each i . If one condition fails, the true equilibrium is a corner solution in which the high-cost firm is inactive and the low-cost firm acts as a monopolist (see Section 4.4).

Finally, total output when both firms are active is:

$$Q^* = \frac{2 - c_1 - c_2}{3 + \alpha}. \quad (38)$$

4.3. Pre-Investment Benchmark (Symmetric Costs c)

The pre-investment benchmark is the symmetric Cournot duopoly with common ownership α and marginal cost c , derived in Section 2. All equilibrium values (equations (6)–(11)) carry over. For convenience, we restate the key result needed for the investment analysis: the pre-investment managerial profit is

$$\Phi_1(\text{pre}) = (1 + \alpha)^2(1 - c)^2 / (3 + \alpha)^2. \quad (39)$$

This value serves as the benchmark against which the post-investment payoff is compared in Section 4.5 below. The remaining pre-investment equations (6)–(11) are referenced directly by their original numbers.

4.4. Post-Investment Equilibrium (Asymmetric Costs $c_1 = 0, c_2 = (1-m)c$)

Suppose firm 1 has invested, so $c_1 = 0$ and $c_2 = (1 - m)c$. Substituting into equation (37) yields the post-investment equilibrium quantities:

$$q_1^{post}(\alpha) = \frac{(1-\alpha) + (1+\alpha)(1-m)c}{(1-\alpha)(3+\alpha)}, \quad (40)$$

$$q_2^{post}(\alpha) = \frac{(1-\alpha) - 2(1-m)c}{(1-\alpha)(3+\alpha)}. \quad (41)$$

Firm 2 produces a positive quantity if and only if its numerator in equation (41) is positive, i.e. $(1-\alpha) - 2(1-m)c > 0$, which gives the exit threshold:

$$\bar{m}(c, \alpha) = \frac{\alpha + 2c - 1}{2c}, \quad \text{or equivalently} \quad \bar{\alpha}(c, m) = 1 - 2c(1 - m). \quad (42)$$

When $m > \bar{m}(c, \alpha)$ (equivalently $\alpha < \bar{\alpha}(c, m)$), both firms remain active. When $m \leq \bar{m}$ (equivalently $\alpha \geq \bar{\alpha}$), firm 2 exits. A stronger spillover m lowers firm 2's cost $(1-m)c$, keeping it competitive and in the market; a higher α works in the opposite direction, because under common ownership the managers value joint profit and joint profit is maximized by concentrating production in the low-cost firm (firm 1, at zero cost). High enough α therefore makes the laggard to withdraw, which is an efficient reallocation motive rather than predatory conduct.

Regime (a): Duopoly ($\alpha < \bar{\alpha}$). Both firms are active. Using the total-output formula (38) with $c_1 = 0$ and $c_2 = (1-m)c$ we obtain:

$$Q^{post}(\alpha) = \frac{2 - (1-m)c}{3+\alpha}, \quad P^{post}(\alpha) = \frac{1+\alpha + (1-m)c}{3+\alpha}. \quad (43)$$

Comparing with the pre-investment benchmark at the same α :

$$Q^{post}(\alpha) - Q_0(\alpha) = \frac{c(1+m)}{3+\alpha} > 0. \quad (44)$$

So, holding α fixed while both firms remain active, the investment always raises total output and therefore consumer surplus. In terms of total output, cost reduction dominates any reallocation effect within this regime.

Regime (b): Monopoly ($\alpha \geq \bar{\alpha}$). Firm 2 is inactive. Firm 1 maximises $\Phi_I = \pi_I$ (since $\pi_2 = 0$) and the monopoly solution gives

$$q^{mono} = \frac{1}{2}, \quad P^{mono} = \frac{1}{2}, \quad \pi^{mono} = \frac{1}{4}, \quad \Phi^{mono} = \frac{1}{4}. \quad (45)$$

Firm 2 optimally stays out because, at $q_I = 1/2$, its best response evaluated at $\alpha \geq \bar{\alpha}$ is non-positive: internalizing a fraction α of firm 1's profit, the laggard does not wish to steal business from its commonly-owned rival.

4.5. The Investment Incentive

Firm 1 invests if and only if the gross gain in its manager's objective, $\Delta\Phi_I = \Phi_I(\text{post}) - \Phi_I(\text{pre})$, is at least the sunk cost F_I . The relevant payoff is the manager's objective Φ_I rather than the raw profit π_I , because it is Φ_I that the manager is contracted to maximize; optimizing π_I instead would deviate from the shareholders' instruction to internalize $\alpha\pi_2$.

The pre-investment value is $\Phi_I(\text{pre}) = (1 + \alpha)^2(1 - c)^2/(3 + \alpha)^2$ from equation (39).

Regime (a): Duopoly ($\alpha < \bar{\alpha}$). In this regime both firms are active and firm 1's post-investment objective is

$$\Phi_1^{\text{post}} = \pi_1^{\text{post}} + \alpha\pi_2^{\text{post}}. \quad (46)$$

The gross gain $\Delta\Phi_I = \Phi_I(\text{post}) - (1 + \alpha)^2(1 - c)^2/(3 + \alpha)^2$ can be written as

$$\Delta\Phi_1(c, m, \alpha) = \frac{N(\alpha, c, m)}{(3 + \alpha)^2(1 - \alpha)} \quad (47)$$

where the numerator N is a quadratic in the spillover parameter m :

$$\begin{aligned} N(\alpha, c, m) &= (2\alpha^2 + 5\alpha + 1)c^2m^2 \\ &- [2(2\alpha^2 + 5\alpha + 1)c^2 + (\alpha^3 - 3\alpha + 2)c] \cdot m \\ &+ (\alpha^3 + 3\alpha^2 + 4\alpha)c^2 - (\alpha^3 + 2\alpha^2 + \alpha - 4)c. \end{aligned}$$

The sign of $\Delta\Phi_I$'s dependence on α reflects two opposing channels:

- **business-stealing channel.** Firm 1 internalizes a fraction α of the profit loss it imposes on firm 2 by becoming more efficient and stealing market share. This partial internalization reduces the private gain from investing.
- **internalized spillover channel.** The investment also reduces firm 2's cost to $(1 - m)c$. Common ownership means firm 1's manager values this cost saving partly as its own, since it raises π_2 and therefore $\Phi_I = \pi_I + \alpha\pi_2$. This strengthens the investment incentive.

This is the López-Vives (2019) logic: whether common ownership strengthens or weakens the incentive to invest depends on the relative size of these two channels, and in particular on the

spillover m . In our model, however, the business-stealing channel never dominates when both firms remain active: throughout the duopoly regime ($\alpha < \bar{\alpha}$), $\Delta\Phi_I(c, m, \alpha)$ is increasing in α , so common ownership always strengthens firm 1's incentive to invest as long as firm 2 stays in the market. The non-monotonicity of $\Delta\Phi_I(\alpha)$ over the full range $[0, 1)$ therefore does not come from a reversal between the two channels within the duopoly regime; it comes entirely from the regime change at $\bar{\alpha}$. Once α is high enough that firm 1's investment drives firm 2 out and firm 1 becomes a monopolist, further increases in α buy no additional business-stealing gain, since the rival is already gone, while they continue to make not investing look more attractive by comparison, so the incentive to invest turns from increasing to decreasing in α exactly at the point where the market tips into monopoly. Firm 1 invests in regime (a) if and only if

$$\Delta\Phi_1(c, m, \alpha) \geq F_I.$$

Regime (b): Monopoly ($\alpha \geq \bar{\alpha}$). Here $\Phi_I(\text{post}) = 1/4$ (equation (45)), so the gross gain simplifies to

$$\Delta\Phi_1(c, \alpha) = \frac{1}{4} - \frac{(1+\alpha)^2(1-c)^2}{(3+\alpha)^2} \quad (48)$$

$\Delta\Phi_I$ is always positive in this regime: since $(1+\alpha)^2(1-c)^2/(3+\alpha)^2 \leq (1-c)^2/4 < 1/4$ for all $c \in (0, 1)$, the monopoly payoff always exceeds the pre-investment duopoly objective, so investment is always profitable in this regime. However, $\Delta\Phi_I$ is decreasing in α : as α rises, the pre-investment payoff $\Phi_I(\text{pre})$ rises while the monopoly payoff $1/4$ is fixed. This means a higher α makes the no-investment outcome more attractive (because common ownership softens competition even without investing), reducing the marginal gain from the investment. Firm 1 invests in regime (b) if and only if $\Delta\Phi_I(c, \alpha) \geq F_I$.

Proposition 3. Firm 1 invests if and only if $\Delta\Phi_1(c, m, \alpha) \geq F_I$, where $\Delta\Phi_1$ is given by equation (47) in the duopoly regime and by equation (48) in the monopoly regime. The effect of common ownership on the investment incentive is non-monotone over the full range $[0, 1)$, and this non-monotonicity is driven by the change in regime at the exit threshold $\bar{\alpha}$: within the duopoly regime ($\alpha < \bar{\alpha}$) the incentive to invest is increasing in α , and it turns decreasing once $\alpha \geq \bar{\alpha}$ and the investment has driven the rival out of the market.

Numerical illustrations, splicing regime (a) below the exit threshold $\bar{\alpha}$ and regime (b) above it:

Example 1, in which $c = 1/4$ and $m = 1/2$.

For these parameter values, the threshold is $\bar{\alpha} = 3/4$. For $\alpha \in [0, 3/4]$:

$$\Delta\Phi_1(1/4, 1/2, \alpha) = \frac{20\alpha^3 + 26\alpha^2 - 9\alpha - 45}{64(\alpha - 1)(\alpha + 3)^2} \quad (49)$$

for $\alpha \in [3/4, 1)$, $\Delta\Phi_1 = 1/4 - 9(1 + \alpha)^2/[16(3 + \alpha)^2]$.

Figure 3 illustrates the results obtained in Example 1:

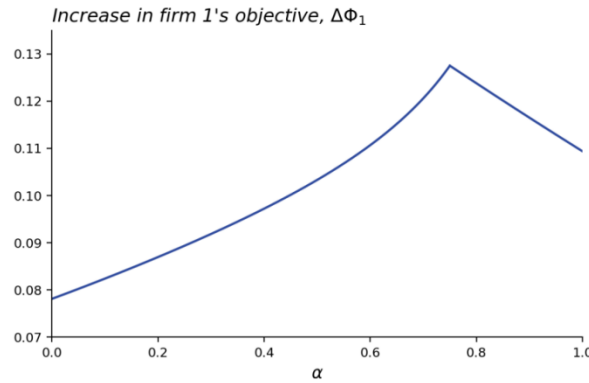


Figure 3. Investment incentive. $\Delta\Phi_1(\alpha)$ for $c = 1/4$, $m = 1/2$ (exit threshold $\bar{\alpha} = 3/4$). $\Delta\Phi_1$ rises from $5/64 \approx 0.078$ at $\alpha = 0$ to $51/400 \approx 0.128$ at $\alpha = 3/4$, then falls to $7/64 \approx 0.109$ at $\alpha = 1$.

Example 2, in which $c = 1/5$, $m = 1/3$:

For these parameter values, the threshold is $\bar{\alpha} = 11/15$. For $\alpha \in [0, 11/15]$:

$$\Delta\Phi_1(1/5, 1/3, \alpha) = \frac{51\alpha^3 + 73\alpha^2 - 11\alpha - 145}{225(\alpha - 1)(\alpha + 3)^2} \quad (50)$$

for $\alpha \in [11/15, 1)$, $\Delta\Phi_1 = 1/4 - 16(1 + \alpha)^2/[25(3 + \alpha)^2]$.

Figure 4 illustrates the results obtained in Example 2:

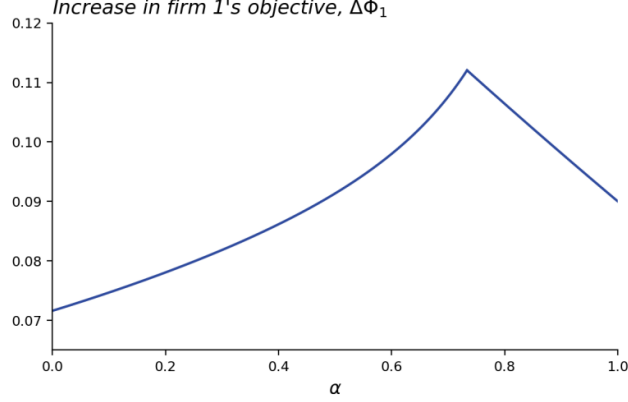


Figure 4. Investment incentive. $\Delta\Phi_1(\alpha)$ for $c = 1/5$, $m = 1/3$ (exit threshold $\bar{\alpha} = 11/15$). $\Delta\Phi_1$ rises from approximately 0.072 at $\alpha = 0$ to approximately 0.110 at $\alpha = 11/15$, then falls toward a positive limit as $\alpha \rightarrow 1$.

In both examples $\Delta\Phi_I$ is first increasing in α in the duopoly regime, then decreasing after the exit threshold. and is positive for all α , so a sufficiently small sunk cost F_I makes the investment worthwhile; the gain is largest at high α , precisely where the investment also monopolizes the market. The investment incentive is therefore non-monotone over the full range $[0, 1)$: increasing in the duopoly regime, then decreasing after the exit threshold in the monopoly regime.

4.6. Consumers, Welfare, and Comparison

Output. Combining the two regimes using the exit threshold $\bar{\alpha}(c, m)$, the post-investment total output as a function of α is

$$Q^{\text{post}}(\alpha) = \begin{cases} \frac{2 - (1 - m)c}{3 + \alpha}, & \alpha < \bar{\alpha} \quad [\text{firm 2 active}] \\ \frac{1}{2}, & \alpha \geq \bar{\alpha} \quad [\text{firm 2 exits}] \end{cases}$$

In the duopoly regime, $Q^{\text{post}}(\alpha)$ falls smoothly in α . At $\alpha = \bar{\alpha}$ firm 2 exits and total output drops continuously to the monopoly level $1/2$. Beyond $\bar{\alpha}$ output is pinned at $1/2$ regardless of α , since firm 1 is a pure monopolist. And we recall the pre-investment output $Q_0(\alpha)$ defined by equation (6).

Effect on consumers. At any fixed α , the investment raises total output in the duopoly regime by $c(1 + m)/(3 + \alpha) > 0$ (equation (44)), so consumer surplus rises. In the monopoly regime, $Q^{\text{post}} =$

$1/2 \geq Q_0(\alpha)$ for all $\alpha \geq \bar{\alpha}$ (since $\bar{\alpha} > 1 - 4c$, which ensures $Q_0(\alpha) < 1/2$ at the exit threshold), so consumers always weakly gain from the investment.

However, a different comparison reveals an ambiguity. When the investment drives firm 2 out (high α , low m), total output settles at $1/2$. This monopoly output need not exceed the competitive benchmark $Q_0(0) = 2(1 - c)/3$: indeed $Q_0(0) \leq 1/2 \Leftrightarrow c \geq 1/4$. For $c < 1/4$, pure competition without innovation delivers more output than the monopolized post-investment equilibrium. Common ownership can thus convert an efficiency gain into a monopolization device that leaves consumers worse off than under standard Cournot competition.

Formally, in the duopoly regime the investment raises consumer surplus relative to the competitive benchmark $Q_0(0)$ if and only if $Q^{\text{post}}(\alpha) \geq Q_0(0)$. Since $Q^{\text{post}}(\alpha) - Q_0(0) = c(1 + m) / (3 + \alpha) - 2\alpha(1 - c) / [3(3 + \alpha)]$, this holds if and only if $\alpha \leq \alpha' = 3c(1 + m) / [2(1 - c)]$. Comparing α' with the exit threshold $\bar{\alpha}(c, m) = 1 - 2c(1 - m)$ shows that $\alpha' < \bar{\alpha}$ exactly when $c < 1/4$. In the monopoly regime, consumers benefit relative to $Q_0(0)$ if and only if $c \geq 1/4$. The region (α, m) where the investment raises consumer surplus relative to $Q_0(0)$ is therefore characterized by $\alpha \leq \alpha'(c, m)$ in the duopoly regime, or $c \geq 1/4$ in the monopoly regime. When $c < 1/4$, α' falls strictly below $\bar{\alpha}$, so there is a nonempty interval $(\alpha', \bar{\alpha})$ in which firm 2 is still active, meaning the market has not yet become a monopoly, yet consumers are already worse off than under standard Cournot competition without investment. Becoming a monopoly is therefore not necessary for the investment to harm consumers.

Welfare. Total welfare is

$$TW^{\text{post}} = CS^{\text{post}} + \pi_1^{\text{post}} + \pi_2^{\text{post}} - F_I. \quad (51)$$

In the duopoly regime, concentrating production in the zero-cost firm 1 improves productive efficiency (lower average cost), which tends to raise welfare; but firm 2's profit shrinks as its cost advantage narrows and α rises. In the monopoly regime ($TW^{\text{mono}} = 1/8 + 1/4 - F_I = 3/8 - F_I$), total welfare depends on whether the productive efficiency gain exceeds the deadweight loss from monopoly pricing plus the sunk cost F_I .

In summary: firm 1 always gains from the investment (higher efficiency and, in regime (b), monopoly rents). Firm 2 loses as α rises and exits entirely in regime (b). Consumers gain against

the same- α counterfactual but lose relative to the no-common-ownership benchmark once α exceeds $\alpha'(c, m)$ when $c < 1/4$. The net welfare sign depends on c , m , and F_I .

Proposition 4. (Consumer surplus.) Against the same- α no-investment counterfactual, the investment always weakly raises consumer surplus. Against the competitive benchmark $Q_0(0)$: (a) if $c \geq 1/4$, consumers always gain; (b) if $c < 1/4$, consumers gain only for $\alpha \leq \alpha'(c, m) = 3c(1 + m) / [2(1 - c)]$, and are worse off for every $\alpha > \alpha'(c, m)$, including a nonempty interval $(\alpha', \bar{\alpha})$ in which firm 2 is still active and under the monopolized post-investment equilibrium ($\alpha \geq \bar{\alpha}$). Becoming a monopoly is not necessary for this harm to arise.

Proof. Since the denominator $3(3 + \alpha)$ is positive, $Q^{\text{post}}(\alpha) - Q_0(0)$ has the same sign as $N(\alpha) = 3c(1 + m) - 2\alpha(1 - c)$, which is strictly decreasing in α , so $Q^{\text{post}}(\alpha) \geq Q_0(0)$ exactly for $\alpha \leq \alpha' = 3c(1 + m) / [2(1 - c)]$. To locate α' relative to the exit threshold $\bar{\alpha} = 1 - 2c(1 - m)$, evaluate N at $\alpha = \bar{\alpha}$: $N(\bar{\alpha}) = 3c(1 + m) - 2(1 - c)\bar{\alpha} = (4c - 1)(2 - c(1 - m))$, and since $c \in (0, 1)$ and $m \in [0, 1]$, the factor $2 - c(1 - m)$ is always strictly positive, so $N(\bar{\alpha})$ has the sign of $4c - 1$. Now, because $N(\alpha)$ is strictly decreasing, $N(\bar{\alpha}) \geq 0$ if and only if $\bar{\alpha} \leq \alpha'$. Equivalently, $\alpha' \geq \bar{\alpha}$ exactly when $c \geq 1/4$, and $\alpha' < \bar{\alpha}$ exactly when $c < 1/4$. Therefore, whenever $c < 1/4$, there exists a nonempty interval $(\alpha', \bar{\alpha})$ in which firm 2 remains active (since $\alpha < \bar{\alpha}$) but consumers are already worse off than under the standard Cournot benchmark. $\square$

Numerical illustrations.

For Example 1 above, figure 5 shows Total output $Q(\alpha)$ before and after innovation:

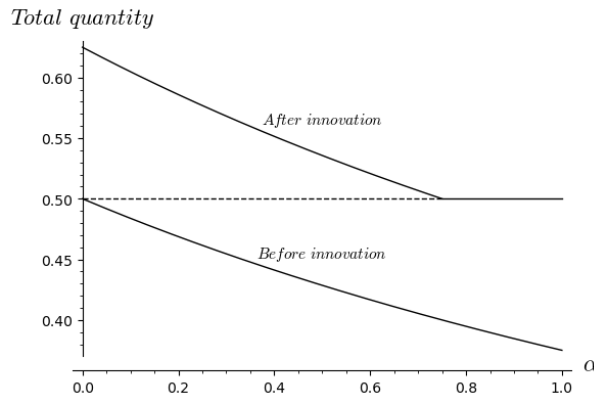


Figure 5. Total output $Q(\alpha)$ before and after innovation for $c = 1/4$, $m = 1/2$ (exit threshold $\bar{\alpha} = 3/4$). The lower curve shows the pre-investment benchmark $Q_0(\alpha) = 2(1-c)/(3+\alpha)$; the upper curve shows the post-investment output $Q^{\text{post}}(\alpha)$, which equals $(2-(1-m)c)/(3+\alpha)$ in the duopoly regime ($\alpha < 3/4$) and drops to the monopoly level $1/2$ once firm 2 exits ($\alpha \geq 3/4$). The dashed line marks $Q = 1/2$. Since $Q_0(0) = 2(1-c)/3 = 1/2$ exactly at $c = 1/4$, the monopolized post-investment outcome delivers the same output as the competitive no-innovation benchmark: this is the boundary case of Proposition 4, where $\alpha' = \bar{\alpha} = 3/4$.

Figure 6 illustrates Total output $Q(\alpha)$ before and after innovation for Example 2 above:

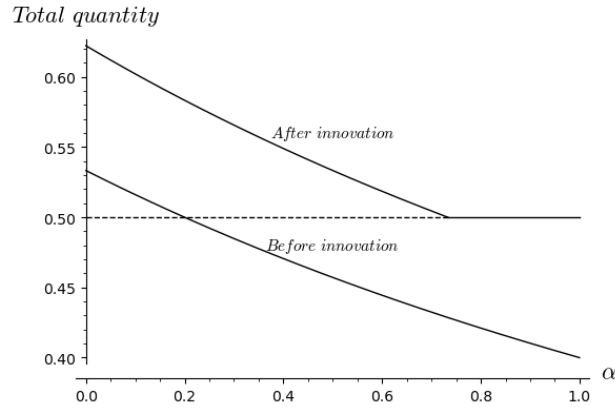


Figure 6. Total output $Q(\alpha)$ before and after innovation for $c = 1/5$, $m = 1/3$ (exit threshold $\bar{\alpha} = 11/15$). The lower curve shows $Q_0(\alpha)$; the upper curve shows $Q^{\text{post}}(\alpha)$, which flattens at $1/2$ once firm 2 exits. Since $c = 1/5 < 1/4$, the competitive benchmark $Q_0(0) = 8/15 \approx 0.533$ exceeds the monopoly output $1/2$ (dashed line), illustrating Proposition 4(b): the upper curve crosses below $Q_0(0)$ already at $\alpha' = 1/2$, well before the kink at the exit threshold $\bar{\alpha} = 11/15$, so consumers are worse off than under standard Cournot competition without innovation over the whole interval $(\alpha', 1)$, not only once firm 2 has exited.

5. Conclusion

This thesis studies how common ownership among competing firms affects entry and innovation, two margins that most existing models hold fixed. Building a two-stage Cournot model in which two incumbents internalize a fraction α of each other's profit, we show that the competitive consequences of common ownership depend critically on the structure of ownership and the technology of the industry, not on the level of common ownership alone.

In the entry model, common ownership confined to incumbents *attracts* entry by an independent outsider, because the incumbents' mutual output restraint raises the market price and expands

residual demand. This is the mirror image of Sato and Matsumura (2020) and Newham, Seldeslachts, and Banal-Estañol (2025), where the entrant is inside the ownership network and common ownership deters entry. The direction therefore depends on whom the ownership links connect. Once regimes are combined, total output and consumer surplus trace a sawtooth path in α , falling within each regime but jumping upward at the entry threshold. Consumers always benefit from entry; total welfare improves only when the sunk cost is sufficiently small.

In the innovation model, two opposing channels shape the investment incentive: a dampened business-stealing channel and an internalized spillover channel. When spillovers are weak and common ownership is high, the investment drives the less efficient rival out, converting an efficiency gain into a device for market concentration. Consumers always weakly gain relative to the same α without investment, but for $c < 1/4$ they are worse off than under standard Cournot competition with neither innovation nor common ownership once α exceeds a threshold $\alpha'(c, m)$ that lies strictly below the exit threshold: becoming a monopoly is sufficient for this harm, but not necessary for it.

The two models may appear to point in opposite directions: in Section 3, common ownership encourages entry, whereas in Section 4 it softens competition to the point of eliminating a rival. Yet both outcomes reflect the same underlying mechanism. In each case, α operates through a single channel: incumbents restrict output, prices rise, and residual demand expands. What differs is the identity of the firm at the margin and whether its objective includes α . In the entry model, the marginal firm is an outsider whose objective is unaffected by α . The residual demand ceded by the incumbents translates directly into profit opportunities, inducing entry. In the innovation model, by contrast, the marginal firm is itself a commonly owned incumbent. The same output restraint now provides an incentive to avoid diverting business from a partner whose profits it partially internalizes. Once the investment creates a sufficiently wide cost gap, withdrawal becomes optimal. Common ownership therefore has no inherent directional effect on market structure. It increases the number of firms when the marginal firm lies outside the ownership network, but reduces it when the marginal firm belongs to that network.

Taken together, these results carry three policy implications. First, antitrust authorities should distinguish between common ownership linking incumbents only (which can attract entry) and

common ownership linking the entrant as well (which deters it). Second, the effect on innovation depends on spillover strength, so policy directed at common ownership in the abstract, without reference to the underlying technology, risks misidentifying both beneficial and harmful cases. Third, industries combining high common ownership with weak spillovers deserve particular scrutiny, since these are precisely the conditions under which innovation concentrates the market rather than benefits consumers.

The analysis in this Master Thesis is subject to several limitations. We discuss now what we see as the most prominent ones:

- First, the model is static and deterministic: it abstracts from repeated interaction, uncertainty over demand or costs, and the possibility that the degree of common ownership α itself responds over time to firms' competitive choices or to regulatory intervention.
- Secondly, throughout the Master Thesis α is treated as an exogenous parameter common to both incumbents, and no analysis is made of the corporate-governance mechanisms (voting, engagement, managerial compensation, etc.), through which common ownership is actually implemented in practice, as well as the possibility that different shareholders exert different degrees of influence.
- Also, the two extensions of the duopoly model with common ownership are studied separately: the entry model abstracts from innovation, and the innovation model abstracts from entry, so the Master Thesis does not characterize how common ownership shapes entry and innovation when they are both present simultaneously.
- Finally, both models retain the stylized features of linear demand, homogeneous products, and Cournot competition in quantities, and the innovation model further restricts attention to a single potential innovator.

These limitations point to avenues for future research. The most natural extension is to combine entry and innovation in the duopoly model: allowing a potential entrant to arrive with its own (possibly superior) technology would show how the entry-attracting effect of common ownership documented in Section 3 interacts with the market-concentrating effect of innovation documented in Section 4. We conjecture that the following could happen: if the entrant brings a superior technology, common ownership among incumbents may simultaneously attract this efficient

entrant (by ceding residual demand) and encourage the incumbent to invest defensively (to avoid being undercut). The net welfare effect would then depend on whether the entrant's technology complements or substitutes for the incumbent's innovation, and on whether common ownership accelerates or delays the entrant's arrival. Such a model would nest both the pro-entry result of Section 3 and the market-concentration result of Section 4 within a single framework and would clarify whether the two effects reinforce or offset each other.

Within the innovation model alone, allowing both incumbents to choose whether to invest would turn Stage 1 into a genuine investment game in which α shapes strategic complementarity or substitutability between the firms' R&D decisions.

Extending the entry framework to allow common ownership between an incumbent and the entrant would nest both this thesis's pro-entry result and the business-stealing-internalization result of Sato and Matsumura (2020) within a single model.

Finally, relaxing the assumptions of homogeneous products and quantity competition, moving to price competition with differentiated products, would test whether these findings extend more broadly.

Summing up, this Master Thesis demonstrates that the competitive consequences of common ownership are not determined by its magnitude alone but by its structure. Whether common ownership helps or harms entry depends on whose ownership is common; whether it helps or harms innovation depends on the strength of technological spillovers. In both cases, the very mechanism that raises prices and softens competition among incumbents can, under the right conditions, expand the market by attracting entry or improve efficiency by financing cost-reducing investment. An approach without accounting for these structural and technological dimensions to common ownership risks missing both the cases in which intervention is warranted and the cases in which it is not.

References

- Antón, Miguel, Florian Ederer, Mireia Giné, and Martin Schmalz. 2023. “Common Ownership, Competition, and Top Management Incentives.” *Journal of Political Economy* 131 (5): 1294–1355.
- Antón, Miguel, Florian Ederer, Mireia Giné, and Martin Schmalz. 2025a. “Innovation: The Bright Side of Common Ownership?” *Management Science* 71 (5): 3713–3733.
- Antón, Miguel, Florian Ederer, Mireia Giné, and Guillermo Ramirez-Chiang. 2025b. “Common Ownership Around the World.” NBER Working Paper No. 33965. Cambridge, MA: National Bureau of Economic Research.
- Azar, José, Sahil Raina, and Martin Schmalz. 2022. “Ultimate Ownership and Bank Competition.” *Financial Management* 51 (1): 227–269.
- Azar, José, Martin Schmalz, and Isabel Tecu. 2018. “Anticompetitive Effects of Common Ownership.” *Journal of Finance* 73 (4): 1513–1565.
- Boller, Lysle, and Fiona Scott Morton. 2020. “Testing the Theory of Common Stock Ownership.” NBER Working Paper No. 27515.
- Borochin, Paul, Jie Yang, and Rongrong Zhang. 2020. “Common Ownership Types and Their Effects on Innovation and Competition.” SSRN Working Paper.
- Bresnahan, Timothy F., and Peter C. Reiss. 1991. “Entry and Competition in Concentrated Markets.” *Journal of Political Economy* 99 (5): 977–1009.
- Bulow, Jeremy I., John D. Geanakoplos, and Paul D. Klemperer. 1985. “Multimarket Oligopoly: Strategic Substitutes and Complements.” *Journal of Political Economy* 93 (3): 488–511.
- Chiao, Cheng-Huei, Bin Qiu, and Bin Wang. 2021. “Corporate Innovation in a World of Common Ownership.” *Managerial Finance* 47 (2): 145–166.
- Cournot, Augustin. 1838. *Recherches sur les Principes Mathématiques de la Théorie des Richesses*. Paris: Hachette.
- Dennis, Patrick, Kristopher Gerardi, and Carola Schenone. 2022. “Common Ownership Does Not Have Anticompetitive Effects in the Airline Industry.” *Journal of Finance* 77 (5): 2765–2798.
- Ederer, Florian, and Isabel Tecu. 2025. “Common Ownership: A Guide for Antitrust Practitioners.”

- In *Antitrust Economics for Lawyers*, chap. 7. LexisNexis.
- He, Jie, and Jiekun Huang. 2017. “Product Market Competition in a World of Cross-Ownership: Evidence from Institutional Blockholdings.” *Review of Financial Studies* 30 (8): 2674–2718.
- Kostovetsky, Leonard, and Alberto Manconi. 2020. “Common Institutional Ownership and Diffusion of Innovation.” SSRN Working Paper.
- Li, Xuelin, Tong Liu, and Lucian A. Taylor. 2023. “Common Ownership and Innovation Efficiency.” *Journal of Financial Economics* 147 (3): 475–497.
- López, Ángel L., and Xavier Vives. 2019. “Overlapping Ownership, R&D Spillovers, and Antitrust Policy.” *Journal of Political Economy* 127 (5): 2394–2437.
- Newham, Melissa, Jo Seldeslachts, and Albert Banal-Estañol. 2025. “Common Ownership and Market Entry: Evidence from the Pharmaceutical Industry.” *American Economic Journal: Microeconomics* 17 (4): 260–327.
- Rotemberg, Julio. 1984. “Financial Transaction Costs and Industrial Performance.” MIT Sloan Working Paper.
- Sato, Susumu, and Toshihiro Matsumura. 2020. “Free Entry under Common Ownership.” *Economics Letters* 195: 109489.
- Singh, Nirvikar, and Xavier Vives. 1984. “Price and Quantity Competition in a Differentiated Duopoly.” *RAND Journal of Economics* 15 (4): 546–554.
- Tirole, Jean. 1988. *The Theory of Industrial Organization*. Cambridge, MA: MIT Press.
- U.S. Department of Justice and Federal Trade Commission. 2023. *Merger Guidelines*. https://www.ftc.gov/system/files/ftc_gov/pdf/2023_merger_guidelines_final_12.18.2023.pdf
- Vives, Xavier. 1999. *Oligopoly Pricing: Old Ideas and New Tools*. Cambridge, MA: MIT Press.